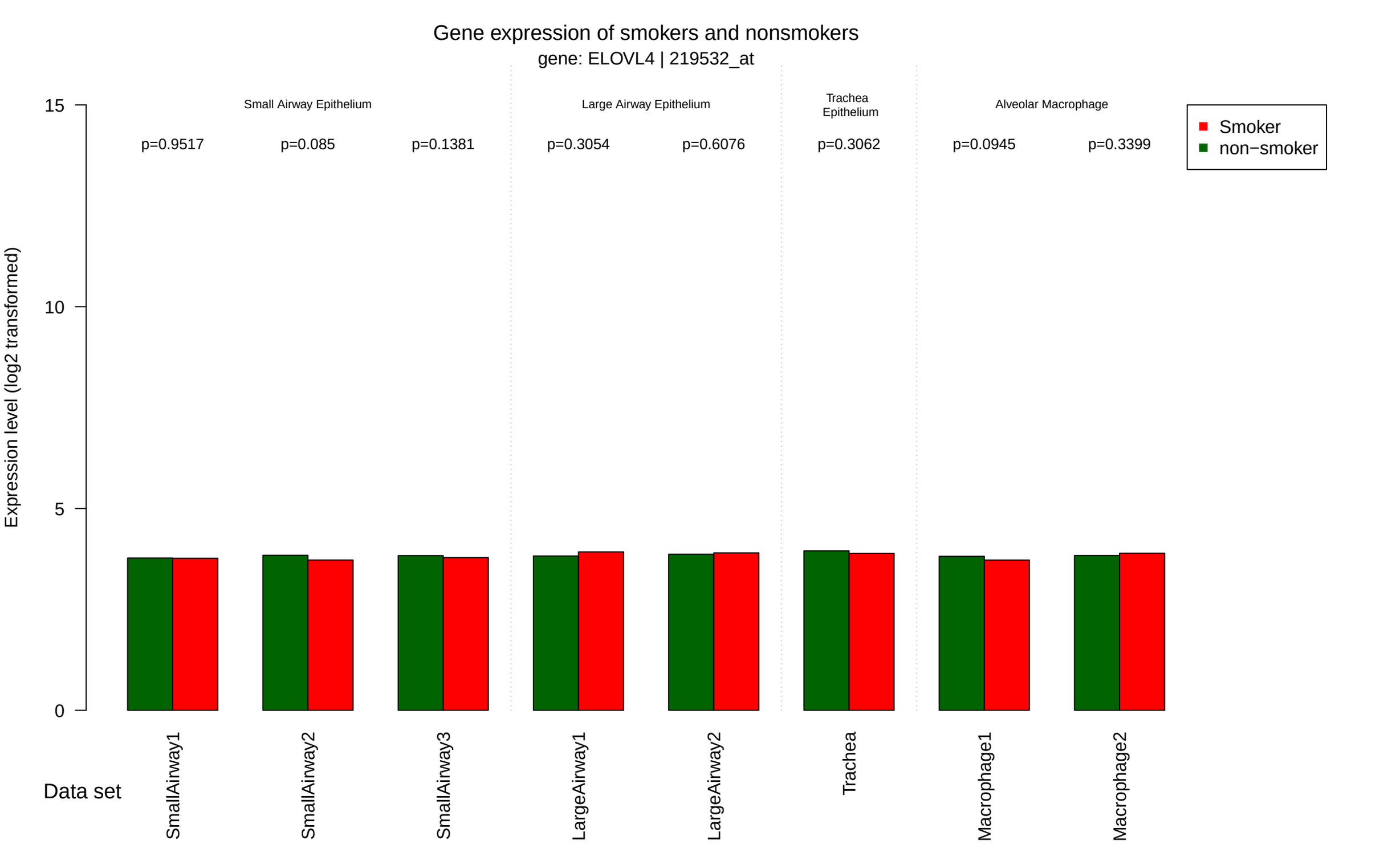
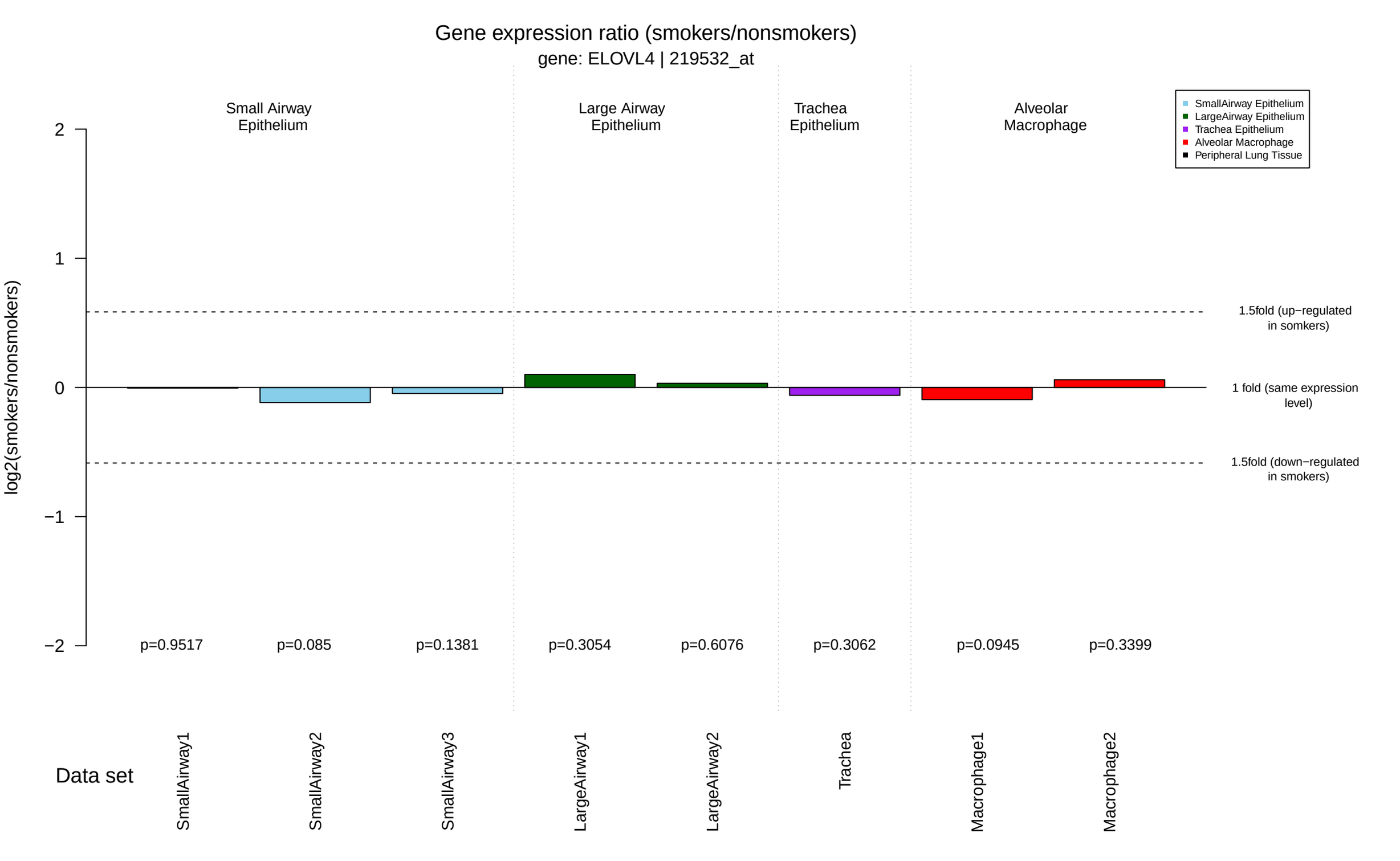
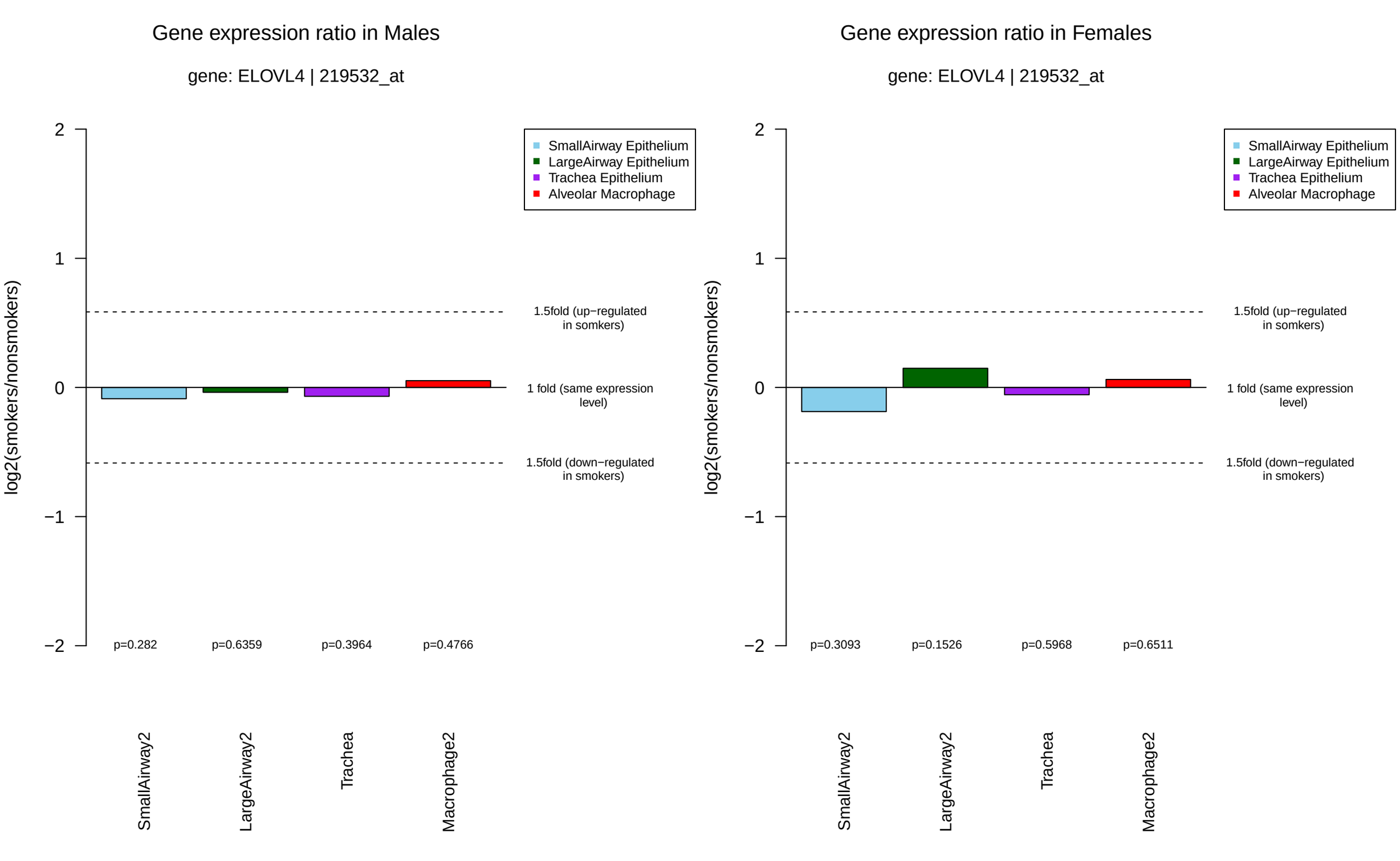
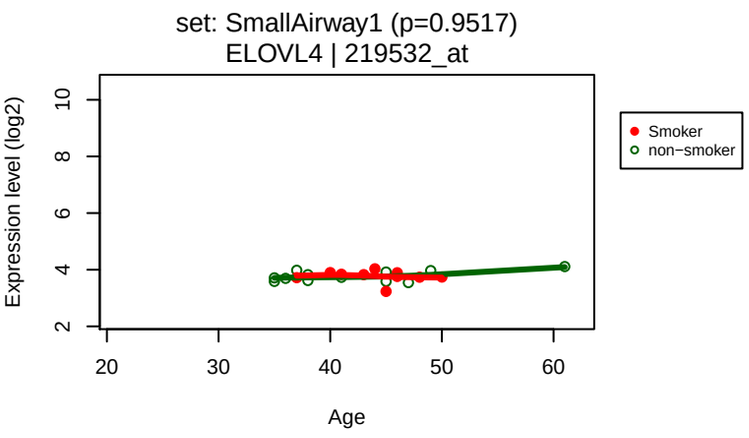
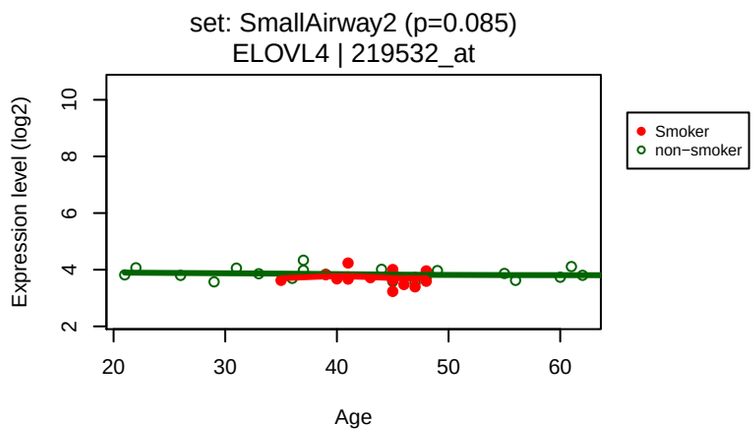
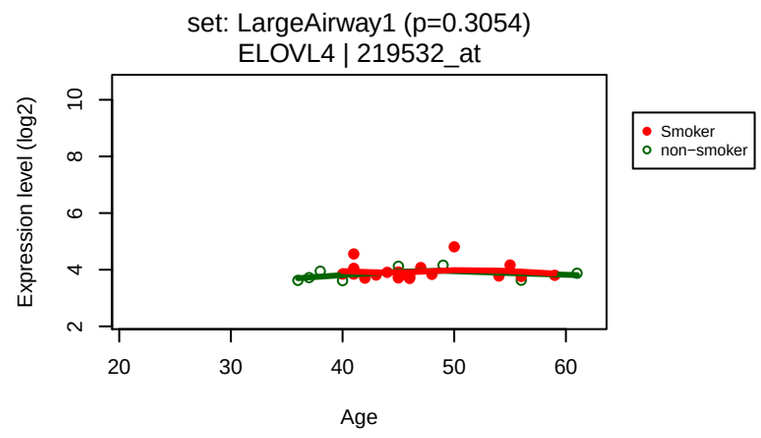

Gene expression of smokers and nonsmokers
gene: ELOVL4 | 219532_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9517
p=0.085
p=0.1381
p=0.3054
p=0.6076
p=0.3062
p=0.0945
p=0.3399
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

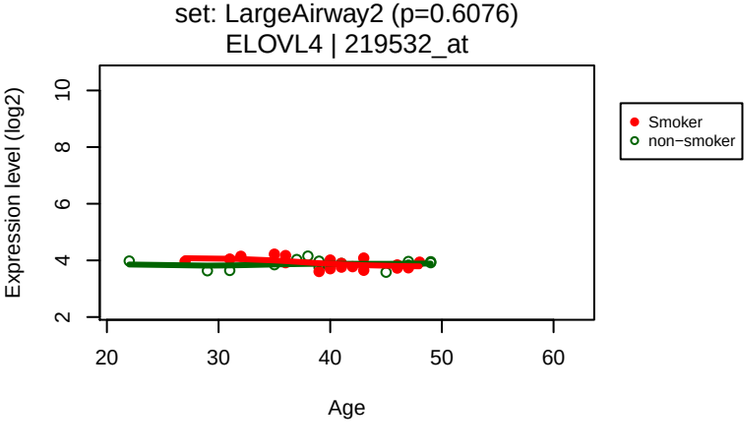

Gene expression ratio (smokers/nonsmokers)
gene: ELOVL4 | 219532_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9517
p=0.085
p=0.1381
p=0.3054
p=0.6076
p=0.3062
p=0.0945
p=0.3399
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

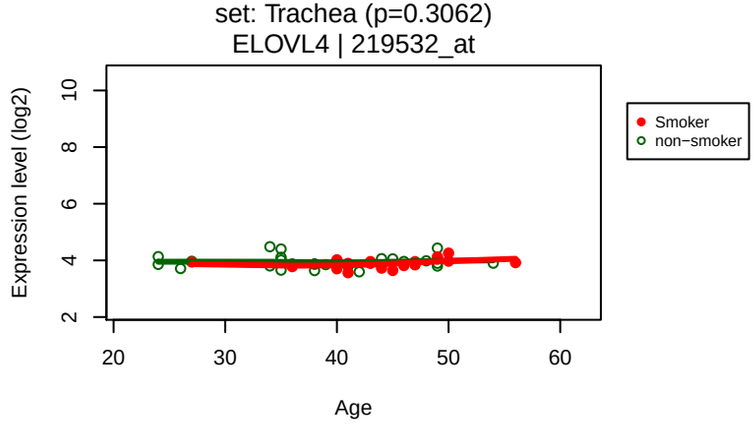

Gene expression ratio in Males
gene: ELOVL4 | 219532_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.282
p=0.6359
p=0.3964
p=0.4766
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ELOVL4 | 219532_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3093
p=0.1526
p=0.5968
p=0.6511
SmallAirway2
LargeAirway2
Trachea
Macrophage2

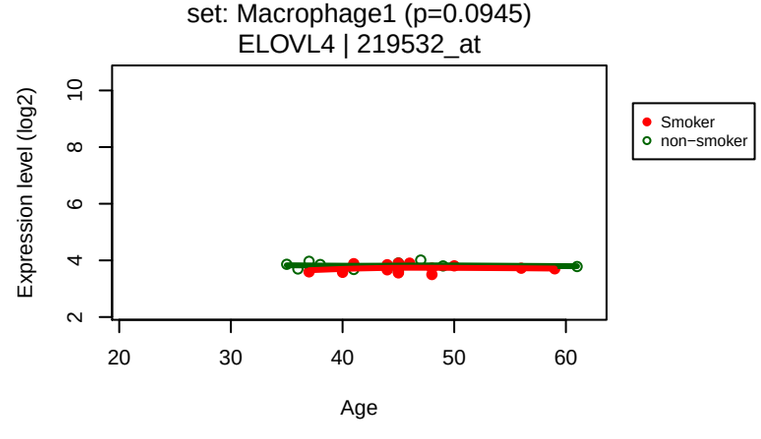

set: SmallAirway1 (p=0.9517)
ELOVL4 | 219532_at
Expression level (log2)
Age
Smoker
non−smoker

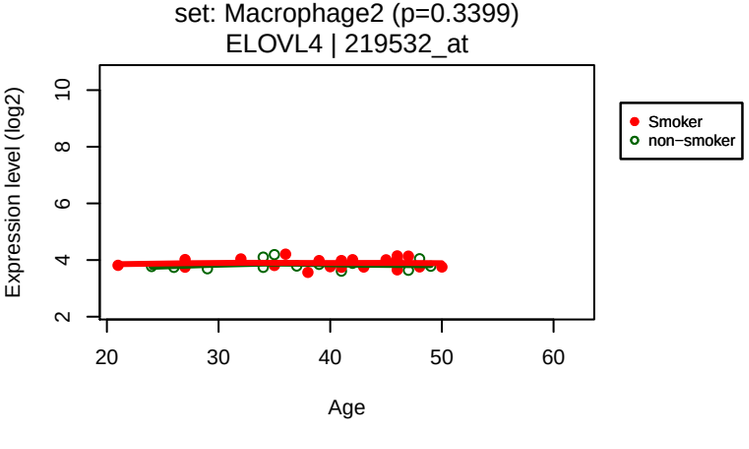

set: SmallAirway2 (p=0.085)
ELOVL4 | 219532_at
Expression level (log2)
Age
Smoker
non−smoker

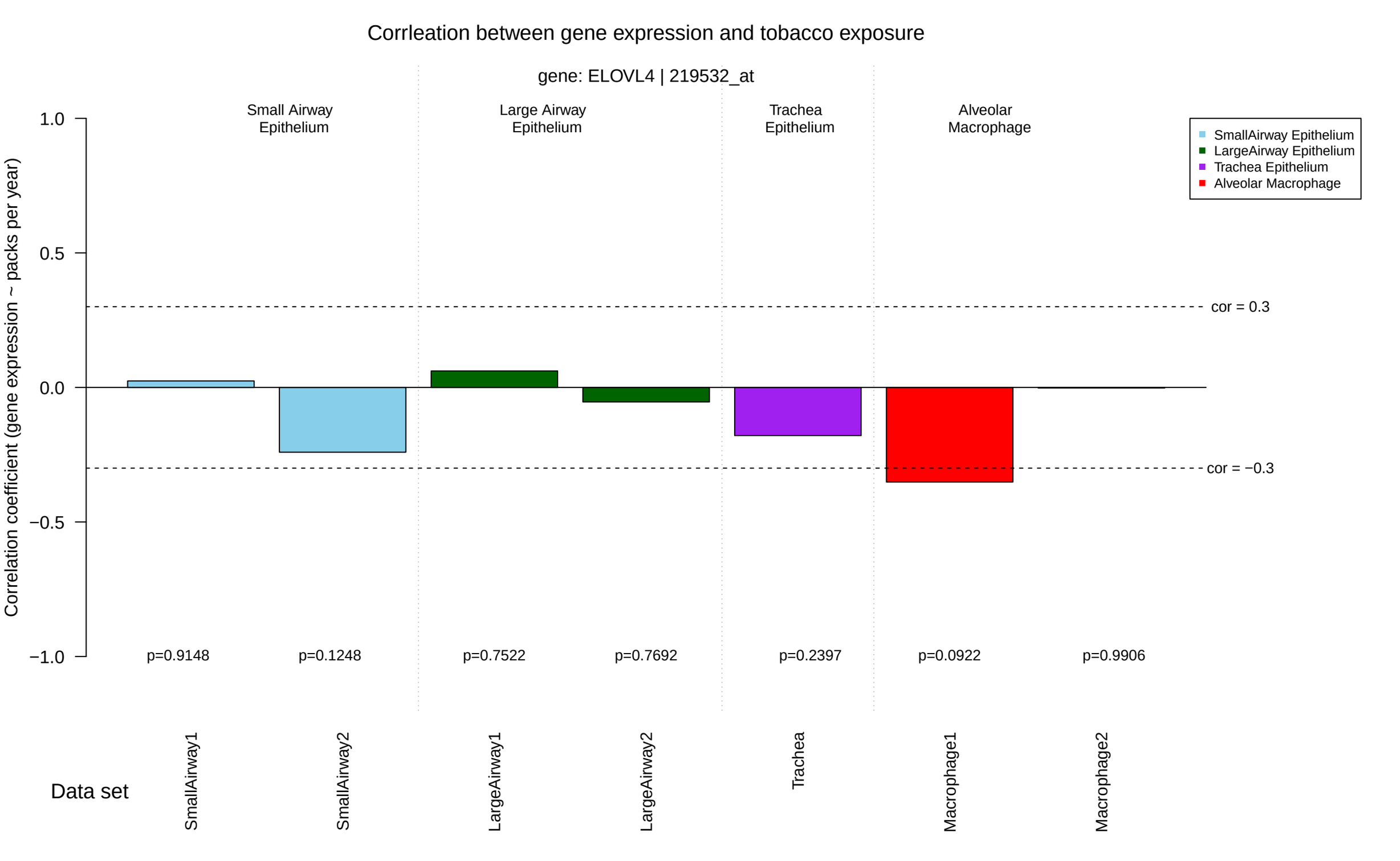

set: LargeAirway1 (p=0.3054)
ELOVL4 | 219532_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6076)
ELOVL4 | 219532_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3062)
ELOVL4 | 219532_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0945)
ELOVL4 | 219532_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3399)
ELOVL4 | 219532_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ELOVL4 | 219532_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9148
p=0.1248
p=0.7522
p=0.7692
p=0.2397
p=0.0922
p=0.9906
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2